

Joonsuk Bae

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ALICE Run 3 charged-particle jets — end-to-end pipeline for ALICE's first Run 3 inclusive charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (preliminary; paper in preparation; EPS-HEP 2025 jet working group merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics — Sungkyunkwan University 2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE*. Defense expected 2026.12; degree conferral 2027.02.

B.S., Physics — Sungkyunkwan University 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN, Geneva	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Research Experience

Main analyzer, ALICE Run 3 charged-particle jets (jet working group, PWG-JE) — Sungkyunkwan University / CERN 2023.04 – present

- **First ALICE Run 3 inclusive charged-particle jet cross section** at $\sqrt{s} = 13.6$ TeV — responsible for the full chain: O²Physics task development, QA on continuous-readout data, Bayesian/SVD unfolding, and the systematic-uncertainty program. ALICE Preliminary; paper in preparation. Varied $R = 0.2$ – 0.6 to separate collinear from large-angle radiation.
- **Tracking-efficiency systematics for Hard Probes 2026** (ALICE jet working group) — derived the pp charged-particle tracking-efficiency uncertainty at $\sqrt{s} = 13.6$ and 5.36 TeV (p_T -binned, 2022–2024 periods), combining the track-selection component with the ITS–TPC matching component.
- **Cross-section normalization** — implemented the Run 3 charged-jet normalization from the ALICE TVX visible cross section under continuous readout, quantifying the bunch-crossing and track-matching inefficiencies introduced by time-frame-based asynchronous reconstruction and deriving the corresponding signal-loss corrections. Now adopted across four Run 3 charged-jet analyses in the group (pp and O+O).
- **Track momentum resolution** — studied data–MC resolution mismatch by covariance propagation, MC residuals, and V0 mass resolution, and derived the p_T -dependent smearing profile used in Run 3 charged-jet systematics.
- **Event-generator production for model comparison** — managed multi-setup POWHEG-BOX and PYTHIA 8 productions (Monash, colour-reconnection modes, rope hadronization, string shoving) plus HERWIG 7 samples for the Run 3 jet-radius comparison, with provenance tracking across configurations.

Detector physicist, ePIC Barrel Imaging Calorimeter (BIC) — Korea-BIC group 2024.02 – present

- **Pb/scintillating-fiber prototype test beams** — lead analyst for the CERN PS T10 (2025.07) campaign: energy response, resolution, and linearity. On-site DAQ shifter with equalization and prompt analysis at KEK AF–AR (2025.03); module performance characterization for the first PS T10 campaign (2024.08), published as a co-author.
- **Module QA and assembly** — ran the PMT-module QA for the prototype modules before the test beams (cosmic-ray and source characterization, timing-coincidence checks, optical-coupling stability, per-module acceptance criteria); Pb/scintillating-fiber bundle alignment, polishing, and PMT/SiPM coupling.
- Energy- and η -dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

Selected Publications

INSPIRE: inspirehep.net/authors/2117232 · ORCID: 0009-0008-4806-8019. ALICE Collaboration co-author since 2022.10; ECCE/ePIC and LAMPS collaboration papers are included in the INSPIRE record.

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (submission expected 2026). Thesis paper. [main analyzer]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider” — CERN PS T10 (2024.08) campaign; submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482); arXiv:2604.22647. [co-author; module-construction support and PMT/module performance characterization]
- Y. Hong, J. Bok, G. An, **J. Bae et al.**, “Test-beam performance of the AstroPix silicon sensor for imaging calorimetry,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). [co-author; beam-test DQM shifts and on-site prompt analysis]
- S. Y. Jang, G. An, **J. Bae et al.**, “Performance of dual-readout calorimeters for various absorbers,” *Nucl. Instrum. Meth. A* (2026), DOI 10.1016/j.nima.2026.171598. [co-author; beam-test participation and DAQ shifts]
- G. Cho, S. Kim, **J. Bae et al.**, “Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders,” *J. Subatom. Part. Cosmol.* **3** (2025) 100021. [co-author; module construction and beam-test data taking]
- Y. J. Kim, D. S. Ahn, J. K. Ahn et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682. [co-author; Time-of-Flight detector maintenance and on-site checks]
- Y. Bae et al. (LAMPS), “Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV,” *Nucl. Instrum. Meth. A* (2025), DOI 10.1016/j.nima.2025.170827. [co-author; Start-Counter prototype testing]
- Y. Cheon et al. (LAMPS), “Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON,” *Nucl. Instrum. Meth. A* **1066** (2024) 169610. [co-author; TPC prototype characterization]
- H. Kim et al. (LAMPS), “Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams,” *JINST* **19** (2024) P12008; arXiv:2412.08662. [co-author; forward-tracking R&D and prototype performance studies]

Invited and Contributed Talks

Summary since 2022: 3 invited workshops [external] · 5 invited collaboration-meeting talks [internal] · 8 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025; a parallel talk and a poster at Hard Probes 2026) · 9 domestic (KPS) · regular speaker in the ALICE jet working group since 2023.

Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

Collaboration-Meeting and National-Workshop Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O^2 MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

Contributed Talks and Posters — International Conferences

- **Hard Probes 2026**, Nashville — *Jet-like dihadron correlations in pO, OO, and Ne-Ne collisions with ALICE* [talk] — 2026.06
- **Hard Probes 2026**, Nashville — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions* [poster] — 2026.06
- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* [on behalf of the ALICE Collaboration; jet working group merge talk] — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE* [talk] — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* [poster] — 2025.04
- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* [poster] — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* [poster] — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC* [poster] — 2023.04

Contributed Talks and Posters — Korean Physical Society

- **KPS Spring 2026**, Daejeon [talk] — *R-dependent charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2026.04
- **KPS Fall 2025**, Daejeon [talk] — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.10
- **KPS Spring 2025** [talk] — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* — 2025.04
- **KPS Spring 2025** [poster] — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** [talk] — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** [talk] — *Charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV* — 2024.04
- **KPS Fall 2023** [talk] — *First look at charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** [poster] — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** [poster] — *Dijet studies with ALICE at the LHC* — 2022.10

Test-beam Activities

ePIC Barrel Imaging Calorimeter test beams

- **CERN PS T10** (2025.07) — lead analyst; energy response and resolution.
- **KEK AF-AR** (2025.03) — DAQ, equalization, prompt analysis.
- **CERN PS T10** (2024.08) — first campaign; module performance characterization.
- **Korea-BIC module production** (2024 – 2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA on the prototype modules used in the 2025 beam-test campaigns.

Service and Professional Activities

- **Tracking-efficiency systematics** for the ALICE jet working group, Hard Probes 2026 (pp 13.6 and 5.36 TeV).
- **ALICE Run 3 O+O charged-jet measurement** — Paper Committee member; co-author on the Analysis Note; derived the cross-section normalization factor and its QA; co-mentor of the MSc main analyzer.
- **ePIC Korea-BIC beam-test analysis framework** — developed and shared the analysis framework across the Korean BIC institutes (SKKU, PNU, KNU).
- **ALICE ITS2 alignment monitoring** (Run 3) — ALICE service task: period-by-period tracking-performance diagnostics from compressed-time-frame (CTF) data.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks.

Awards and Fellowships

- **BK21 FOUR Graduate Scholarship** — awarded on research performance by the Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.

Teaching and Mentoring

Student mentoring (non-supervisory) — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., main analyzer of the ALICE Run 3 O+O charged-jet measurement) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author on the corresponding ALICE Analysis Note.
- *Harim Seong* (undergraduate) — ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

KoALICE O² Tutorial — student co-organizer and instructor — Sungkyunkwan University, Suwon 2024.01.22–23

- Co-organized (with H. Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community, ~22 participants across Korean institutes; instructor on *Running O² Physics with Hyperloop*.

Teaching Assistant, SKKU Physics (2022 – 2024) — *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I*.

Technical Skills

Languages. English (professional working proficiency), Korean (native).

Programming. C++, Python, ROOT, L^AT_EX, Bash, Git.

HEP frameworks. ALICE O²Physics, Geant4, JANA2 / EIC-recon, FastJet.

Software contributions. ALICE O²Physics — jet analysis tasks and shared framework tools.

Event generators. PYTHIA 8 (Monash, colour-reconnection modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.

Analysis methods. jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

Detector hardware. PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

References

Available on request.